CHAPTER

# 04

# Memory and Forgetting

Do you remember what you ate for breakfast, the name of your first teacher, or how to ride a bicycle? Without memory, life would have no continuity: we could not use past skills, recognize family members, or learn anything. Learning and memory describe roughly the same processes: **learning** usually refers to the **first acquisition** (encoding) of information, while **memory** refers to its later **storage and retrieval**. This chapter explains how memory works, its stages and types, why we forget, and how to remember better, which you can apply directly to your own studying.

LESSONS IN THIS CHAPTER


**4.1** The Meaning and Processes of Memory

**4.2** The Stages of Memory: The Atkinson–Shiffrin Model

**4.3** Types of Long-Term Memory, the Serial Position Effect and Memory Factors

**4.4** Forgetting and Its Theories

**4.5** Improving Memory


Freshman A+ · General Psychology (Psyc 1011)

CHAPTER 4 · LESSON 4.1

# The Meaning and Processes of Memory



**AFTER THIS LESSON YOU CAN**

- Define memory
- Explain encoding, storage and retrieval in order
- Locate the cause of a memory failure in the right process
- Distinguish recall, recognition and relearning

## 4.1.1 What memory is

**D MEMORY**

**Memory** is the **retention of information over time**: the way we record the past for later use in the present. It is not one single process but a collection of abilities and processes. Memory is also **not like a video camera**: memories can change and be distorted as they are stored and retrieved.

## 4.1.2 The three processes of memory

**F ENCODING → STORAGE → RETRIEVAL**

| Process | Meaning | Computer comparison | Example |
|---|---|---|---|
| **1. Encoding** | **putting information in**: transforming sensory input into a **code** (form) memory can use. The term comes from **computer science**. | typing on a keyboard | reading a definition and understanding its meaning |
| **2. Storage** | **keeping** the information: the encoded experience leaves a **memory trace** in the nervous system and is held over time | saving the file on a disk | the definition stays in your memory until the exam |
| **3. Retrieval** | **getting information out**: locating it, bringing it into awareness and using it | opening the file | writing the definition in the exam |

**★ A MEMORY CAN FAIL AT ANY OF THE THREE STAGES**

- **Encoding failure:** if you encode a new word only as a **sound**, there is no memory trace of its **meaning**. You cannot recall what was never encoded.
- **Storage failure:** the trace fades or is lost.
- **Retrieval failure:** the item is **still stored** but was "**misfiled**", so you cannot find it now (e.g. a lecturer who cannot recall the name of a student he taught for three years). **Information cannot be retrieved unless it was properly encoded and stored first.**

**◎ THE THREE STAGES IN A STUDENT'S EXAM**

Kedir reads his notes while chatting on his phone, so the definition of "chunking" never really enters memory (**encoding failure**). Selam studies it carefully but never reviews it for two months, and it fades (**storage** problem). Meron knows it well, but in the exam hall her mind goes blank, then she remembers it on the way home (**retrieval failure**).

## 4.1.3 Measuring memory: recall, recognition and relearning

| Method | What you do | Example |
|---|---|---|
| **Recall** | reproduce the information **without** cues (or with few) | essay and fill-in-the-blank questions |
| **Recognition** | **identify** the correct information among choices | multiple-choice and true/false questions |
| **Relearning (savings)** | learn the material again and measure **how much faster** it is the second time | re-studying last semester's course takes less time than the first time |

Recognition is usually easier than recall. Relearning, Ebbinghaus's method, shows that "forgotten" material still leaves traces.

## Exam Practice

10 QUESTIONS · LESSON 4.1

**1** Debre Birhan University · Final 2015

_____ is the process of receiving sensory input and transforming it into a code that can be stored.

A) Decoding
B) Retrieval
C) Storage
D) Encoding

**2** Wolkite University · Final 2015

Which arrangement presents the three memory processes in the order in which they occur?

A) encoding, storage, retrieval
B) encoding, retrieval, storage
C) storage, retrieval, encoding
D) storage, encoding, retrieval

**3** Dilla University · Mid 2015

Placing information into memory is called **__, while getting information out of memory is called ____**.

A) storage; encoding
B) storage; retrieval
C) encoding; retrieval
D) encoding; decoding

**4** Wolkite University · Final 2015

The step in the memory process that actually makes our memories available to us is:

A) retrieval
B) encoding
C) rehearsal
D) storage

**5** Addis Ababa University · Mid 2014

Which option correctly matches the memory terms with their meanings?

A) Retrieving = converting information; encoding = storing; storing = holding
B) Storing = converting; encoding = holding; retrieving = getting out
C) Encoding = converting information; storing = holding information; retrieving = getting information out
D) Encoding = getting out; storing = converting; retrieving = holding

**6** Arsi University · Final 2014

Information cannot be retrieved unless it was properly encoded and stored in the first place.

True / False

**7** Wolkite University · Final 2015

Memory is a single process that enables us to retrieve and use stored information.

True / False

**8** Arba Minch University · Final 2016

Walking down the street, Dr. Ayele bumped into a former student he had taught for three years, but he could not remember the student's name. Later that day it came to him. This is an example of:

**A)** proactive interference
**B)** retroactive interference
**C)** retrieval failure
**D)** motivated forgetting

**9** Madda Walabu University · Final 2014

The recall method measures memory by the time it takes to relearn forgotten material.

True / False

**10** Model question (module)

A student reads a new term but pays attention only to how it sounds, not to what it means. In the exam she cannot remember its meaning. At which memory process did the failure occur?

**A)** Encoding
**B)** Storage
**C)** Retrieval
**D)** Sensory adaptation

## Answers & Explanations

**1.** D

**Encoding**

Encoding transforms input into a usable memory code.

**2.** A

**encoding, storage, retrieval**

Information must be put in, kept, and then taken out.

**3.** C

**encoding; retrieval**

Encoding puts information in; retrieval takes it out.

**4.** A

**retrieval**

Retrieval locates stored information and brings it into awareness.

**5.** C

**Encoding = converting information; storing = holding information; retrieving = getting information out**

Encoding converts input into a memory code, storage holds it, retrieval brings it out.

**6.** True

Retrieval depends on successful encoding and storage.

**7.** False

Memory involves three processes (encoding, storage, retrieval) and several structures.

**8.** C

**retrieval failure**

The name was stored (it came back later) but could not be accessed at that moment.

**9.** False

That describes the relearning (savings) method. Recall means reproducing information without cues.

**10.** A

**Encoding**

The meaning was never encoded, so there is no memory trace of it to store or retrieve.

CHAPTER 4 · LESSON 4.2

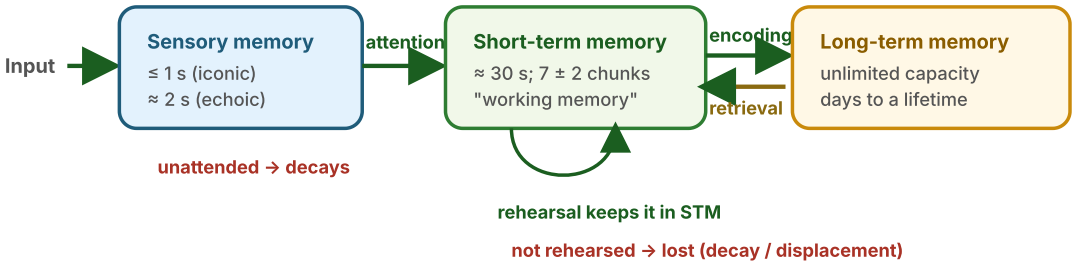
# The Stages of Memory: The Atkinson–Shiffrin Model

**AFTER THIS LESSON YOU CAN**

- Describe the information-processing (three-box) model
- Compare sensory, short-term and long-term memory in capacity and duration
- Explain iconic and echoic memory
- Explain chunking and rehearsal

## 4.2.1 The three-box model

Cognitive psychologists compare the mind to a **computer** that takes information in, processes it in stages and produces output. The most influential model is that of **Richard Atkinson and Richard Shiffrin (1968)**: information flows through **three memory stores**.




*The Atkinson–Shiffrin three-stage model of memory*

## 4.2.2 Sensory memory (sensory register)

The **entry point** of memory. It holds **almost all** the information reaching the senses, as an accurate but **unprocessed** copy, for a very **brief** time, giving us a moment to decide what is important. There is a separate register for each sense:

- **Iconic memory** = **visual** images, up to about **1 second**.
- **Echoic memory** = **auditory** (sound) images, up to about **2 seconds**.
  Most of it simply **decays**; only what receives **attention** passes on to short-term memory.

## 4.2.3 Short-term memory (STM, working memory)

STM holds the **contents of our attention**: the results of perceptual analysis that we are thinking about now. It is used for thinking, reading, speaking and problem solving, so it is also called **working**, immediate, active or primary memory.

**F FOUR CHARACTERISTICS OF STM**

| Characteristic | Meaning |
|---|---|
| **Active** | information stays only while you are consciously working on it; a mental "workspace" |
| **Rapid accessibility** | information is instantly available (a file on top of your desk vs one in a drawer) |
| **Preserves sequence** | keeps the order of items (a phone number, the steps of a calculation) |
| **Limited capacity** | George **Miller (1956)**: "the magic number **7 ± 2**", i.e. about 5 to 9 items, or more precisely **7 ± 2 chunks** |
| **Duration:** about **30 seconds** unless rehearsed. | |

**D CHUNKING AND REHEARSAL**

- **Chunking:** grouping small pieces of information into **larger meaningful units** (chunks). It increases the **capacity** of STM (more information per chunk), **not** its duration. Example: 0 9 1 1 2 3 4 5 6 7 is hard; **0911 · 234 · 567** is easy. "ETHIOPIAN" is 9 letters but 1 chunk.
- **Rehearsal:** consciously repeating information, which keeps it in STM indefinitely.
- **Maintenance rehearsal:** simple **repetition** (saying a phone number over and over; reading a handout again and again without understanding). It keeps information in STM but transfers it poorly to LTM.
- **Elaborative rehearsal:** linking new information to what you **already know** and to its **meaning** (remembering a new person called Ali because your brother is also called Ali). It is much better for storing information in LTM.

## 4.2.4 Long-term memory (LTM)

The store for **relatively permanent**, meaningful information. Its capacity has **no practical limit**, and it lasts days, months, years or a lifetime. It holds our knowledge, skills and personal history (our sense of identity). Information moves from STM to LTM through **encoding**, especially elaborative rehearsal, and comes back to STM through **retrieval**.

**F THE THREE STORES COMPARED**

| | Sensory memory | Short-term memory | Long-term memory |
|---|---|---|---|
| Capacity | very large (almost everything sensed) | small: 7 ± 2 chunks | practically unlimited |
| Duration | ≤ 1 s (visual), ≈ 2 s (sound) | ≈ 30 s (longer with rehearsal) | days to a lifetime |
| Form | unprocessed sensory copy | results of perception: words, images, sounds | meaningful, organized |
| Information is lost by | decay | decay, displacement, interference | interference, retrieval failure, motivated forgetting |

**STAGE TRAPS**

- Sensory memory is the **briefest**, not the most permanent.
- Sensory memory does **not** do the mental "work"; manipulation for thinking and learning happens in **STM** (working memory).
- STM capacity = **7 ± 2 chunks**, not bits.
- Chunking increases STM **capacity**, not duration. Rehearsal increases duration.
- Repeating material without understanding is **maintenance** rehearsal; linking it to meaning is **elaborative** rehearsal.

## Exam Practice

14 QUESTIONS · LESSON 4.2

**1** Addis Ababa University · Mid 2014

In Atkinson and Shiffrin's model, a fast-decaying store of visual information is called __, **and a fast-decaying store of auditory information is called** ____.

**A)** semantic memory; episodic memory
**B)** echoic memory; iconic memory
**C)** iconic memory; echoic memory
**D)** episodic memory; semantic memory

**2** Debre Birhan University · Final 2015

Auditory information stored in sensory memory is called:

**A)** iconic memory
**B)** echoic memory
**C)** non-declarative memory
**D)** episodic memory

**3** Dilla University · Mid 2015

Information in the sensory register is more permanent than information in short-term memory.

True / False

**4** Arba Minch University · Final 2015

Sensory memory stores information for manipulation in cognitive tasks such as learning.

True / False

**5** Debre Birhan University · Final 2015

Short-term memory stores about five to nine items of information.

True / False

**6** Wolkite University · Final 2015

What is the actual capacity of short-term memory?

**A)** 7 ± 2 bits of information
**B)** 7 ± 2 chunks of information
**C)** 5 bits of information
**D)** 9 chunks of information

**7** Wolkite University · Final 2015

Chunking is the memory procedure that involves:

**A)** discarding some items to remember the rest better
**B)** suppressing short-term memory altogether
**C)** grouping information into meaningful units
**D)** repeating information many times

**8** Madda Walabu University · Final 2014

Chunking increases the duration of information in short-term memory.

True / False

**9** Debre Birhan University · Final 2015

A good way to keep a friend's phone number in mind is to group the digits. This method is called:

**A)** Chunking
**B)** Maintenance rehearsal
**C)** Elaborative rehearsal
**D)** Organization

**10** Arba Minch University · Final 2016

Saba meets a man named Ali, the same name as her brother, and easily remembers his name by making this association. This is:

**A)** Implicit memory
**B)** Chunking
**C)** Maintenance rehearsal
**D)** Elaborative rehearsal

**11** Jigjiga University · Final (Set A)

A student tries to memorize the psychology handout by reading it again and again without understanding the meaning. This indicates:

**A)** Maintenance rehearsal
**B)** Elaborative rehearsal
**C)** Chunking
**D)** Organization

**12** Debre Birhan University · Final 2015

Elaborative rehearsal involves repeated verbalization or thinking of the material.

True / False

**13** Dilla University · Mid 2015

At a party, Melat is introduced to many family members and repeats each new name as often as she can. These names are being kept in her:

**A)** iconic memory
**B)** long-term memory
**C)** short-term memory
**D)** sensory memory

**14** Jimma University · Final 2014

Which statement about the stages of memory is true?

**A)** Sensory memory lasts a few minutes
**B)** STM lasts as briefly as sensory memory
**C)** STM is brief, easily interrupted and limited in capacity
**D)** LTM has a capacity of 7 ± 2 items

## Answers & Explanations

**1.** **C**

**iconic memory; echoic memory**

Iconic = visual (about 1 s); echoic = auditory (about 2 s).

**2.** **B**

**echoic memory**

"Echo" relates to sound.

3. **False**

Sensory memory lasts only 1–2 seconds; STM lasts about 30 seconds and can be extended by rehearsal.

4. **False**

Mental manipulation happens in short-term (working) memory. Sensory memory only holds raw input briefly.

5. **True**

Miller's "magic number" is 7 ± 2.

6. **B**

**7 ± 2 chunks of information**

Through chunking, STM holds about seven chunks, each of which can contain several bits.

7. **C**

**grouping information into meaningful units**

Chunking packs pieces into larger meaningful units.

8. **False**

Chunking increases how much STM can hold (capacity). Rehearsal increases how long it is held.

9. **A**

**Chunking**

Grouping digits into units (0911-234-567) is chunking.

10. **D**

**Elaborative rehearsal**

Linking new information to existing knowledge is elaborative rehearsal.

11. **A**

**Maintenance rehearsal**

Simple repetition without meaning is maintenance rehearsal.

12. **False**

That is maintenance rehearsal. Elaborative rehearsal connects the material to meaning and prior knowledge.

13. **C**

**short-term memory**

Rehearsal (repetition) keeps information active in short-term memory.

14. **C**

**STM is brief, easily interrupted and limited in capacity**

STM lasts about 30 s, is easily displaced, and holds about 7 chunks. Sensory memory lasts seconds; LTM is unlimited.

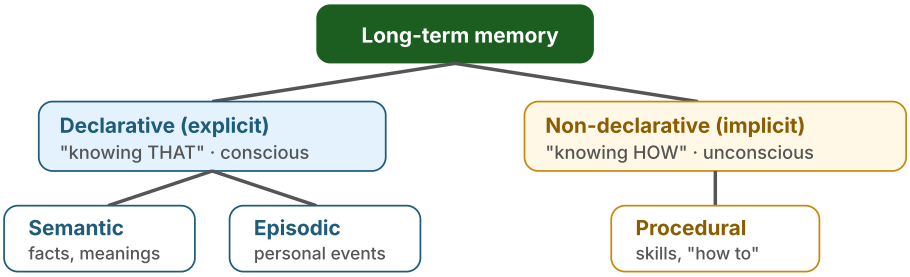
CHAPTER 4 · LESSON 4.3

# Types of Long-Term Memory, the Serial Position Effect and Memory Factors

**AFTER THIS LESSON YOU CAN**

- Distinguish declarative (semantic, episodic) and non-declarative (procedural) memory
- Classify any memory example
- Explain the serial position effect with the three-box model
- List the factors that affect memory

## 4.3.1 How long-term memory is organized




*The subsystems of long-term memory*

**F TYPES OF LONG-TERM MEMORY**

| Type | What it stores | Examples |
|---|---|---|
| **Declarative / explicit** | **conscious** recollection of facts or events that can be **put into words** | |
| → **Semantic** | **general facts and knowledge**, meanings of words, concepts, maths, independent of when and where you learned them | the definition of psychology; Addis Ababa is the capital; a friend called a "walking encyclopedia" |
| → **Episodic** | **personal experiences**: events tied to a specific **time and place** | your last birthday party; how you met your dorm-mate; a fun day at secondary school; your father's childhood stories; remembering where you fell off your bicycle when you see the scar |
| **Non-declarative / implicit** | memories that affect behaviour **without conscious recollection** | |
| → **Procedural** | **skills and habits**: "knowing **how**" | riding a bicycle, swimming, folding laundry, climbing a tree, chopping onions, playing the guitar |

**? CLASSIFYING A MEMORY (WOLAITA SODO 2014)**

**"Imote defines 'psychology' exactly as she learned it two years ago." Which memory?**

Step 1: Is it a **skill** ("how to")? No → declarative.

Step 2: Is it tied to a **personal event** with a time and place (*when I…*)? No, it is a general fact → **semantic**.

If the question asked "Imote remembers the **day** her teacher first defined psychology and where she sat" → **episodic**.

**Quick test:** Fact → semantic. "I remember when I…" → episodic. "I can do…" → procedural.

## 4.3.2 The serial position effect

When you study a list and recall it immediately, recall is best for items at the **beginning** (**primacy effect**) and at the **end** (**recency effect**), and worst in the **middle**: a **U-shaped curve**. At a party you remember the first and last people you met but not those in between.

★ **WHY IT HAPPENS (THREE-BOX MODEL)**

- **Primacy:** STM was "empty" when the first items arrived, so they were rehearsed well and moved into **LTM**.
- **Recency:** the last items are **still in STM** at the moment of recall.
- **Middle items:** STM was already crowded, so many dropped out before reaching LTM.

## 4.3.3 Factors affecting memory

Good memory depends on **learning, retention and remembering**. The module lists: ability to retain (good memory traces); **good health**; **age** (young people remember better than the aged); **maturity** (very young children cannot retain complex material); **will** to remember; **intelligence**; **interest**; **overlearning**; **speed** of learning; **meaningfulness** of material; and **sleep or rest** after learning, which strengthens memory.

! **MEMORY-TYPE TRAPS**

- Declarative memory is "knowing **that**"; "knowing how to do things" is **procedural**, which is **non-declarative**.
- **Recalling a mobile number** is declarative (semantic), not implicit.
- Explicit memory involves **conscious** recollection; implicit memory does not.
- **Serial position** is an effect of recall, not a technique that helps store information in LTM.

## Exam Practice

13 QUESTIONS · LESSON 4.3

**1** Addis Ababa University · Mid 2014

When we recall personal experiences, such as who spoke at a party last weekend or what we did on our last birthday, we are recalling _____ memory.

**A)** semantic
**B)** procedural
**C)** episodic
**D)** non-declarative

**2** Arsi University · Final 2014

Looking at a scar on his knee, Daniel recalls the time and place he fell off his bicycle. What type of memory is this?

**A)** Iconic memory
**B)** Episodic memory
**C)** Procedural memory
**D)** Semantic memory

**3** Madda Walabu University · Final 2014

Bontu, a first-year student, tells her classmates about a colourful party she attended in grade 11. This narration comes from her:

**A)** semantic memory
**B)** procedural memory
**C)** episodic memory
**D)** short-term memory

**4** Madda Walabu University · Final 2014

Gurmessa's friends call him a "walking encyclopedia" because of his general knowledge. His memory is most probably strong in:

**A)** short-term memory
**B)** semantic memory
**C)** procedural memory
**D)** episodic memory

**5** Wolaita Sodo University · Final 2014

Imote defines the term "psychology" exactly as she learned it two years ago. From which long-term memory is she retrieving this information?

**A)** Non-declarative memory
**B)** Implicit memory
**C)** Episodic memory
**D)** Semantic memory

**6** Arba Minch University · Final 2016

Mr. Tesema knows how to ride a bicycle, fold laundry and climb a tree. In which memory is this information stored?

**A)** Semantic memory
**B)** Episodic memory
**C)** Procedural memory
**D)** Explicit memory

**7** Arba Minch University · Final 2015

Which one of the following is NOT an example of implicit memory?

**A)** Chopping an onion
**B)** Riding a motorbike
**C)** Recalling mobile phone numbers
**D)** Playing an electric guitar

**8** Debre Birhan University · Final 2015

All of the following are true about declarative memory EXCEPT:

**A)** It is memory for knowing how to do things
**B)** It is memory for facts, concepts and general knowledge
**C)** It requires conscious effort to recollect
**D)** It includes episodic memory

**9** Arba Minch University · Final 2015

Explicit memory involves the conscious recollection of factual information.

True / False

**10** Arba Minch University · Final 2016

The tendency to recall the last items in a sequence more readily than those in the middle is called the:

**A)** Recency effect
**B)** Primacy effect
**C)** Retrieval
**D)** Recognition

**11** Bahir Dar University · Mid 2014

"People usually remember best the things that come first and last in learning." This principle is equivalent in meaning to:

**A)** primacy and recency
**B)** readiness and primacy
**C)** exercise and readiness
**D)** mood and reward

**12** Madda Walabu University · Final 2014

All of the following help you store information in long-term memory EXCEPT:

**A)** Elaboration
**B)** Organization
**C)** Serial position
**D)** Context

**13** Arba Minch University · Final 2015

Youngsters can remember complex material better than the aged.

True / False

## Answers & Explanations

**1.** C

**episodic**

Personal events tied to a time and place are episodic.

**2.** B

**Episodic memory**

He recalls a personal event with its time and place.

**3.** C

**episodic memory**

It is a personal experience from a specific time and place.

**4.** B

**semantic memory**

General knowledge and facts are semantic memory.

**5.** D

**Semantic memory**

A definition is factual knowledge (semantic), not a personal event.

**6.** C

**Procedural memory**

These are skills ("knowing how").

**7.** C

**Recalling mobile phone numbers**

Recalling numbers is conscious, declarative memory. The others are procedural skills.

**8.** A

**It is memory for knowing how to do things**

"Knowing how" is procedural (non-declarative) memory.

**9.** True

Explicit (declarative) memory is consciously recalled.

**10.** A

**Recency effect**

The last items are still in short-term memory at recall.

**11.** A

**primacy and recency**

First items = primacy; last items = recency (the serial position effect).

**12.** C

**Serial position**

Serial position describes which items are recalled best; it is not a storage strategy.

**13.** True

The module lists age as a factor: young people generally remember better than the elderly (though very young children cannot retain complex material).

CHAPTER 4 · LESSON 4.4

# Forgetting and Its Theories

**AFTER THIS LESSON YOU CAN**

- Define forgetting and describe Ebbinghaus's forgetting curve
- Explain decay, interference, displacement, motivated and cue-dependent forgetting
- Distinguish proactive and retroactive interference
- Apply the theories to real cases

## 4.4.1 What forgetting is

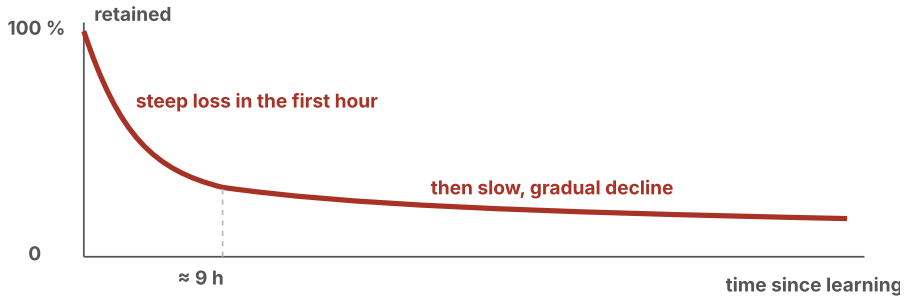
**D FORGETTING**

The **apparent loss of information already encoded and stored** in long-term memory. Forgetting is not always bad: we meet so much information daily that storing all of it would overwhelm us, so we are **selective**. Sometimes we even forget **on purpose** (motivated forgetting).

**S EBBINGHAUS AND THE FORGETTING CURVE**

The German psychologist **Hermann Ebbinghaus (1885)**, using himself as the only subject, memorized lists of **nonsense syllables** (FIW, BOZ) and measured how quickly he could **relearn** them after different delays (the **savings** method).

- Forgetting is **fastest at first**, especially in the **first hour**.
- After about **nine hours** the rate slows, and little more is lost even after many days.
- **Relearning** is almost always faster than learning from scratch.
  **Study lesson:** review new lecture notes the **same day**, before the steepest forgetting.




*Ebbinghaus's forgetting curve*

## 4.4.2 Five theories of forgetting

### F THEORIES OF FORGETTING

| Theory | Cause of forgetting | Best explains | Example |
|---|---|---|---|
| **Decay** | memory traces (engrams) **fade with the passage of time** if not used | sensory memory and STM; weaker for LTM (we forget yesterday but remember childhood) | memories of your grade-one teacher gradually becoming blurred |
| **Interference** | **similar** information gets **confused** in storage or retrieval | LTM | see the table below |
| **Displacement** (new memory for old) | new information **pushes out** old information, like recording over a tape | **STM** (limited to 7 ± 2 chunks), not LTM | forgetting the first digits of a number when someone tells you another number |
| **Motivated forgetting** (repression) | we push **threatening or painful** memories out of consciousness (Freud called it **repression**) | emotional, traumatic memories | a person who cannot recall a painful childhood experience |
| **Cue-dependent** (retrieval failure) | the right **retrieval cues** are missing; the memory is there but cannot be reached | probably the **most common** failure in LTM | remembering better in the **same room** (context) or the **same mood or physical state** in which you learned (**state-dependent** memory) |

### ★ PROACTIVE VS RETROACTIVE INTERFERENCE

| | Direction | Memory affected | Examples |
|---|---|---|---|
| **Proactive** | **old → new**: earlier learning interferes **forward** | the **new** information is hard to recall | writing last year's date in January; writing your **old** phone number or typing your **old** password; calling Debebe "Abebe" because you met Abebe first; maths studied yesterday making today's physics harder to learn |
| **Retroactive** | **new → old**: later learning interferes **backward** | the **old** information is hard to recall | a salesman forgets an earlier customer's transaction because twenty more came after it; after learning your new password, you cannot recall the old one |
| **Memory aid: PRO** = **P**rior **R**uins **O**ne new; **RETRO** = new goes **back** and ruins the old. Ask: *which memory is lost?* If the **new** one is lost → proactive. If the **old** one is lost → retroactive. | | | |

### ? SOLVING AN INTERFERENCE QUESTION (DILLA 2015)

**"Tewodros recently changed his phone number. Now he often writes his old number on forms." Which theory?**
Step 1: Is there **competition between two similar memories**? Yes (two phone numbers) → interference, not decay.

Step 2: Which memory is **failing**? The **new** number.
Step 3: Old interferes with new → **proactive interference**.

**! FORGETTING TRAPS**

- "Decay = losing information because we purposely repress it" is **false**; repression is motivated forgetting. Decay depends on **time**.
- The theory that treats **passage of time** as critical is **decay**.
- **Displacement** applies to **STM**, not LTM.
- Retroactive interference: **new** learning disrupts **old** memories (not the reverse).

## Exam Practice

17 QUESTIONS · LESSON 4.4

**1** Addis Ababa University · Mid 2014

In the first few days of each new year, many of us keep writing the old year in the date. This is mainly due to:

**A)** decay of the memory trace
**B)** proactive interference
**C)** retroactive interference
**D)** motivated forgetting

**2** Dilla University · Mid 2015

Tewodros recently changed his phone number. Now he often writes his old number on forms. This is best explained by:

**A)** cue-dependent forgetting
**B)** retroactive interference
**C)** proactive interference
**D)** decay

**3** Wolaita Sodo University · Final 2014

Mr. Tolosa has just changed his phone password, and he is frustrated because he keeps trying the old password. This is an example of:

**A)** proactive interference
**B)** retroactive interference
**C)** retrograde amnesia
**D)** anterograde amnesia

**4** Bonga University · Mid 2014

At a party you meet someone named Abebe. A little later you meet Debebe, but you call him "Abebe" by mistake. This is most likely:

**A)** retroactive interference
**B)** displacement
**C)** proactive interference
**D)** decay

**5** Arba Minch University · Final 2016

The mathematics I studied yesterday makes it harder to learn and remember the physics I am studying today. Which kind of forgetting is this?

**A)** Decay
**B)** Proactive interference
**C)** Retroactive interference
**D)** Retrieval failure

**6** Arsi University · Final 2014

A salesman forgets an earlier customer's transaction because twenty other transactions were completed after it. His forgetting is caused by:

**A)** decay of information
**B)** lack of a retrieval cue
**C)** proactive interference
**D)** retroactive interference

**7** Wolaita Sodo University · Final 2014

Retroactive interference theorists believe that previously learned information is powerful enough to banish current information.

True / False

**8** Debre Birhan University · Final 2015

Proactive interference occurs when prior information interferes with new information.

True / False

**9** Madda Walabu University · Final 2014

For which theory is the passage of time the significant factor in forgetting?

**A)** Decay
**B)** Interference
**C)** Motivated forgetting
**D)** Reconstruction

**10** Jigjiga University · Final (Set A)

Your memory of your grade-one teacher gradually becomes blurred as you move up through school. This best illustrates which theory of forgetting?

**A)** Decay
**B)** Interference
**C)** Motivated forgetting
**D)** Cue-dependent forgetting

**11** Debre Birhan University · Final 2015

Decay theory states that information is lost when we purposely repress it.

True / False

**12** Dilla University · Mid 2015

Fantahun tries to forget his painful childhood experiences. This is explained by:

**A)** Interference theory
**B)** Motivated forgetting theory
**C)** Displacement theory
**D)** Decay theory

**13** Bonga University · Mid 2014

The theory of forgetting which holds that banishing emotionally threatening or painful events from the conscious mind brings about forgetting is:

**A)** Decay theory
**B)** Motivated forgetting theory
**C)** Interference theory
**D)** Cue-dependent theory

**14** Madda Walabu University · Final 2014

A woman cannot recall a traumatic event from ten years ago because she does not want to remember it. Which theory best describes her forgetting?

**A)** Decay
**B)** Interference
**C)** Motivated forgetting
**D)** Reconstruction

**15** Jigjiga University · Mid 2012

We remember information more easily when the context is similar to the original one. This supports:

A) Displacement theory
B) Motivated forgetting
C) Cue-dependent theory
D) Decay theory

**16** Model question (module)

A student studied for an exam while drinking strong coffee and felt very alert. She remembers the material best when she is again in the same alert state. This is called:

A) state-dependent memory
B) proactive interference
C) decay
D) displacement

**17** Model question (module)

Which theory of forgetting mainly applies to short-term memory because of its limited capacity?

A) Motivated forgetting
B) Displacement (new memory for old)
C) Cue-dependent forgetting
D) Repression

## Answers & Explanations

**1.** B

**proactive interference**

The old (habitual) year interferes with recalling the new one.

**2.** C

**proactive interference**

The old number (prior learning) interferes with the new one.

**3.** A

**proactive interference**

Old learning interferes with recalling the new password.

**4.** C

**proactive interference**

The first name learned interferes with remembering the second.

**5.** B

**Proactive interference**

Earlier learning (maths) interferes with later learning (physics).

**6.** D

**retroactive interference**

The later transactions interfere with memory of the earlier one.

**7.** False

That describes proactive interference. In retroactive interference, new information disrupts old.

**8.** True

Pro = forward: old learning disrupts new.

**9.** A

**Decay**

Decay theory says traces fade with time when not used.

**10.** A

**Decay**

The memory fades gradually through disuse over time.

**11.** False

Purposeful repression is motivated forgetting; decay is fading with time.

**12.** B

**Motivated forgetting theory**

Pushing painful memories out of awareness is motivated forgetting (repression).

**13.** B

**Motivated forgetting theory**

Freud called this repression; today it is called motivated forgetting.

**14.** C

**Motivated forgetting**

The memory is blocked because it is painful, not because of time or similar information.

**15.** C

**Cue-dependent theory**

Cues present at learning help retrieval later.

**16.** A

**state-dependent memory**

Your physical or emotional state acts as a retrieval cue (a type of cue-dependent memory).

**17.** B

**Displacement (new memory for old)**

New items push out old ones in STM, which holds only about 7 chunks. LTM's capacity is practically unlimited.

CHAPTER 4 · LESSON 4.5

# Improving Memory



**AFTER THIS LESSON YOU CAN**

- Apply the module's six guidelines for better memory
- Use mnemonics and distributed practice when studying
- Plan an evidence-based study routine

## 4.5.1 Six guidelines from the module

**HOW TO REMEMBER BETTER**

| Guideline | Why it works | What to do |
|---|---|---|
| **Pay attention** | we often forget because we never **encoded** the information | put the phone away; sit in front |
| **Encode in more than one way** | more elaborate encoding creates more retrieval routes | read it, say it, draw it, write it |
| **Add meaning** | meaningful material links to what is already in LTM | relate concepts to your own life (elaborative rehearsal) |
| **Take your time** | breaks and rest reduce **interference**; **sleep** is the best protection | distributed study, not one night of cramming |
| **Overlearn** | studying beyond the point of "I know it" makes memory durable | keep reviewing after you can recall it once |
| **Monitor your learning** | **testing yourself** shows what you really know | practice questions, flashcards, explaining to a friend |

**A MEMORY-SMART STUDY PLAN FOR A FRESHMAN**

1. **Same day:** rewrite the lecture notes in your own words (beat the steep part of the forgetting curve).
2. **Next day:** make a one-page chart comparing concepts (e.g. the five forgetting theories), adding your own examples (meaning).
3. **Each weekend:** test yourself with practice questions without looking (monitoring, retrieval practice).
4. **Before the exam:** review once more even if you feel ready (overlearning), then **sleep** well.
5. Use **mnemonics** for lists, as in this book: "**A** **R**eal **R**ole **M**odel" for Bandura's four steps.

## Exam Practice

4 QUESTIONS · LESSON 4.5

**1** Arba Minch University · Final 2016

Which of the following is NOT a way to improve memory?

**A)** Paying attention
**B)** Adding meaning
**C)** Increasing interference
**D)** Overlearning

**2** Bonga University · Mid 2014

Which memory-improvement technique involves frequently testing yourself?

**A)** Overlearning
**B)** Adding meaning
**C)** Monitoring learning
**D)** Attention

**3** Model question (module)

A student keeps studying the forgetting theories even after she can recite them perfectly. Which guideline is she using?

**A)** Overlearning
**B)** Paying attention
**C)** Taking breaks
**D)** Chunking

**4** Model question (module)

Why is sleeping after studying recommended?

**A)** It increases proactive interference
**B)** It reduces interference and strengthens memory connections
**C)** It moves information into sensory memory
**D)** It speeds up decay

## Answers & Explanations

**1.** C

**Increasing interference**

Interference causes forgetting; breaks and sleep are recommended to reduce it.

**2.** C

**Monitoring learning**

Testing yourself, rehearsing and reviewing show how well you know the material.

**3.** A

**Overlearning**

Overlearning means continuing to study after you think you know the material.

**4.** B

**It reduces interference and strengthens memory connections**

Sleep and rest after learning reduce interference and consolidate memory.

CHAPTER 4 · QUICK REVISION

# Memory and Forgetting — at a glance

## CHAPTER 4 AT A GLANCE

| Topic | Must-know points |
|---|---|
| Memory | retention of information over time; not a video recording |
| Processes | encoding (in, code) → storage (keep, trace) → retrieval (out); failures at any stage |
| Measures | recall, recognition, relearning (savings) |
| Model | Atkinson & Shiffrin: sensory (iconic ≤ 1 s, echoic ≈ 2 s) → STM (≈ 30 s, 7 ± 2 chunks, working memory) → LTM (unlimited) |
| STM tools | chunking (capacity), maintenance vs elaborative rehearsal (duration / transfer to LTM) |
| LTM types | declarative: semantic (facts), episodic (personal events); non-declarative: procedural (skills) |
| Serial position | primacy (in LTM) + recency (still in STM) = U-shaped curve |
| Forgetting | Ebbinghaus (fast at first, then slow; relearning saves time) |
| Theories | decay (time), interference (proactive: old → new; retroactive: new → old), displacement (STM), motivated (repression), cue-dependent (context, state) |
| Improve | attention, multiple encoding, meaning, breaks and sleep, overlearning, self-testing |

## THREE LTM TYPES: "FACTS, EPISODES, PROCEDURES"

**S**emantic = **S**chool facts. **E**pisodic = **E**vents in **my** life. **P**rocedural = **P**ractised skills.